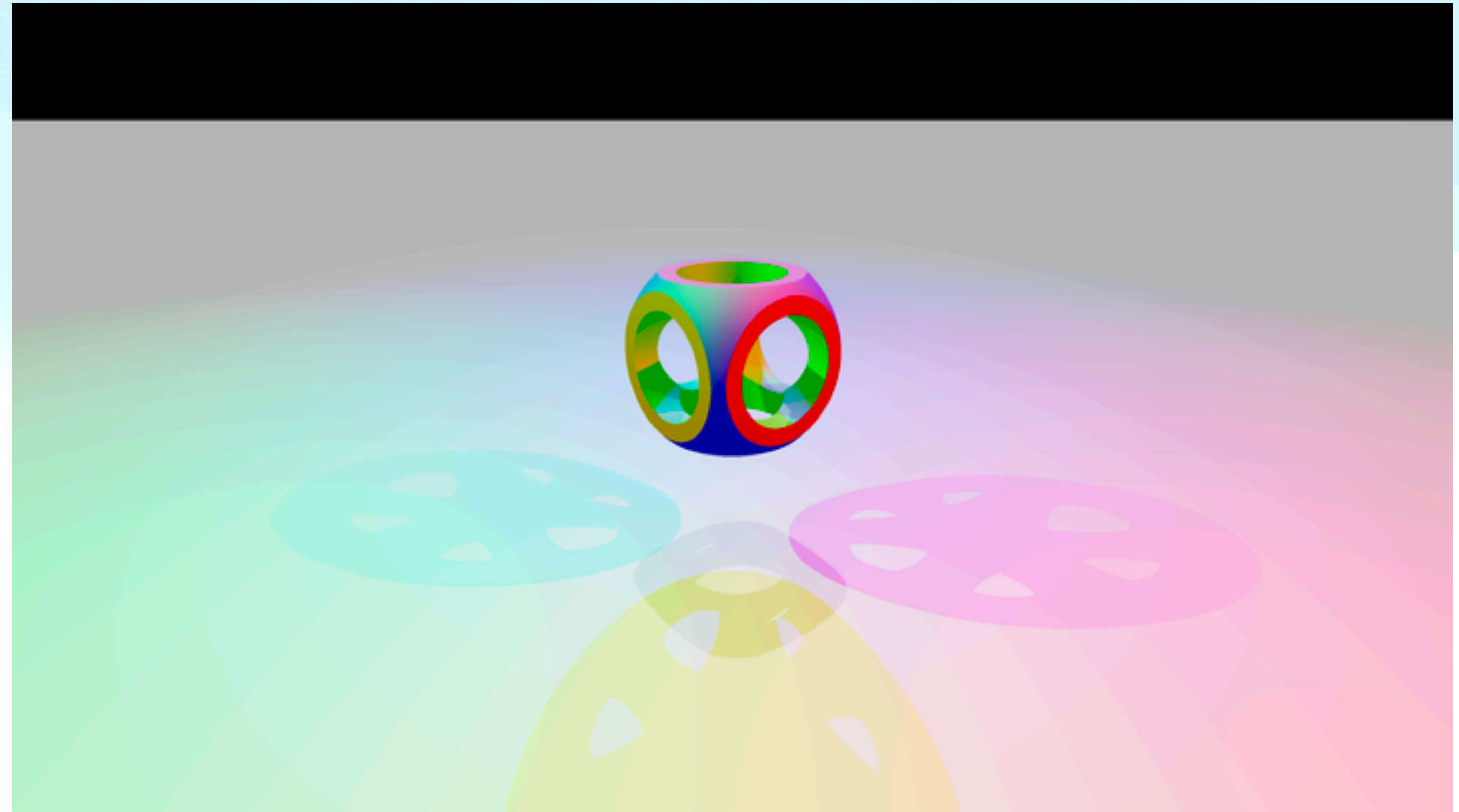


HDR (High Dynamic Range)

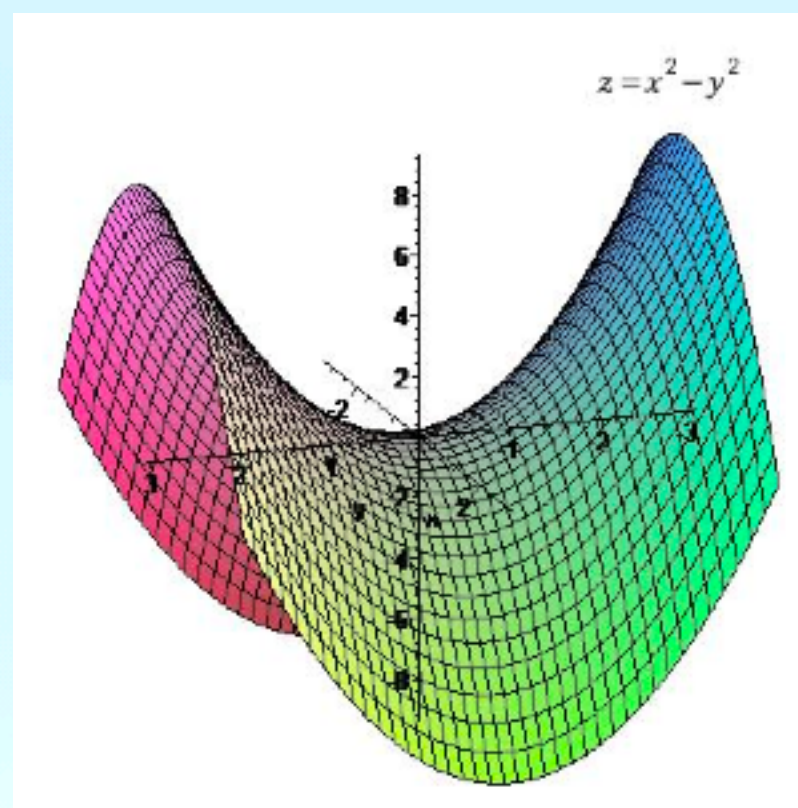
- HDR이 뭐죠?

- WHAT :
이미지 내 밝은 부분과 어두운 차이를 극대화시키는 기법
- WHY :
더 극적인 이미지를 얻을 수 있다.
- HOW :
빛을 $0 \sim \text{infinity}$ 로 계산
-> $0 \sim 1$ 로 비선형적으로 정규화



BONUS minRT에서 해볼만한 것들

- CSG, HDR
- 2차곡면 일반화
- 다차곡면
- 그림자, 반사, 굴절
- multi threading
- global illumination
- obj 파일 파싱 -> 드래곤



$$\begin{aligned}
 C_{xxx} x^4 &= C_p^4 + 4C_{px}^3 + 6C_{pxx}^2 + 4C_{pxx}^2 + C_{xxx}^2 \\
 C_{xxy} y^4 &= C_q^4 + 4C_{qy}^3 + 6C_{qyy}^2 + 4C_{qyy}^2 + C_{yyy}^2 \\
 C_{xyx} z^4 &= C_r^4 + 4C_{rx}^3 + 6C_{rxy}^2 + 4C_{rxy}^2 + C_{xxx}^2 \\
 C_{xyx} xy^4 &= C_{pq}^4 + C_{px}^3 + 3C_{pxy}^2 + 3C_{pxy}^2 + 3C_{pxy}^2 + 3C_{pxy}^2 + C_{xyx}^2 + C_{xyx}^2 \\
 C_{xyx} x^4 &= C_{pr}^4 + C_{py}^3 + 3C_{prx}^2 + 3C_{prx}^2 + 3C_{prx}^2 + 3C_{prx}^2 + C_{xyx}^2 + C_{xyx}^2 \\
 C_{xyx} xy^4 &= C_{pq}^4 + 3C_{pqx}^2 + 3C_{pqx}^2 + C_{py}^3 + C_{qy}^3 + 3C_{qyx}^2 + 3C_{qyx}^2 + C_{xyx}^2 \\
 C_{xyx} y^4 &= C_{qy}^4 + C_{qy}^3 + 3C_{qyr}^2 + 3C_{qyr}^2 + 3C_{qyr}^2 + 3C_{qyr}^2 + C_{xyx}^2 + C_{xyx}^2 \\
 C_{xyx} x^4 &= C_{pr}^4 + 3C_{prx}^2 + 3C_{prx}^2 + C_{py}^3 + C_{rxy}^3 + 3C_{rxy}^2 + 3C_{rxy}^2 + C_{xyx}^2 \\
 C_{xyx} y^4 &= C_{qr}^4 + 3C_{qrx}^2 + 3C_{qrx}^2 + C_{qy}^3 + C_{ry}^3 + 3C_{ryx}^2 + 3C_{ryx}^2 + C_{xyx}^2 \\
 C_{xyx} xy^4 &= C_{pq}^4 + 2C_{pqx}^2 + C_{py}^3 + 2C_{pqx}^2 + 2C_{pqx}^2 + C_{qy}^3 + 2C_{qyx}^2 + C_{xyx}^2 \\
 C_{xyx} y^4 &= C_{qr}^4 + 2C_{qrx}^2 + C_{qy}^3 + 2C_{qrx}^2 + 2C_{qrx}^2 + C_{ry}^3 + 2C_{ryx}^2 + C_{xyx}^2 \\
 C_{xyx} x^4 &= C_{pr}^4 + 2C_{prx}^2 + C_{py}^3 + 2C_{prx}^2 + 2C_{prx}^2 + C_{rxy}^3 + 2C_{rxy}^2 + C_{xyx}^2 \\
 C_{xyx} xy^4 &= C_{pq}^4 + C_{pqx}^2 + C_{py}^3 + 2C_{pqx}^2 + 2C_{pqx}^2 + C_{qy}^3 + 2C_{qyx}^2 + C_{xyx}^2 \\
 C_{xyx} y^4 &= C_{qr}^4 + C_{qrx}^2 + 2C_{qrx}^2 + 2C_{qrx}^2 + C_{py}^3 + C_{qy}^3 + 2C_{qyx}^2 + C_{xyx}^2 \\
 C_{xyx} x^4 &= C_{pr}^4 + 2C_{prx}^2 + C_{py}^3 + 2C_{prx}^2 + C_{py}^3 + C_{rxy}^3 + 2C_{rxy}^2 + C_{xyx}^2 \\
 C_{xyx} y^4 &= C_{qr}^4 + 2C_{qrx}^2 + C_{qy}^3 + 2C_{qrx}^2 + C_{qy}^3 + C_{ryx}^3 + 2C_{ryx}^2 + C_{xyx}^2 \\
 C_{xyx} x^4 &= C_{pr}^4 + 3C_{prx}^2 + 3C_{prx}^2 + C_{py}^3
 \end{aligned}$$

